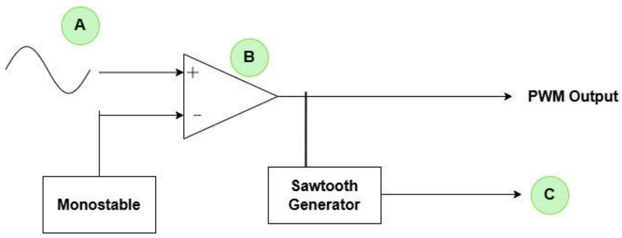
SOMAIYA VIDYAVIHAR UNIVERSITY

Maximum Marks: 50	**Semester: January 2025– April 2025** **Examination: End-Semester Examination**		**Duration: 2 Hrs.**
Programme code: 03 **Programme:** B.Tech Electronics and Telecommunication Engineering		**Class:** SY	**Semester:** IV **(SVU 2023)**
Institute/School/Department: K. J. Somaiya School of Engineering		**Name of the department:** EXTC	
Course Code: 216U03C403	**Name of the Course:** Communication Systems		
Instructions: 1) Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary			

Que. No.	Question Statement	Max. Marks
Q.1	Attempt any **two**	
i)	Suggest the method of transmitting analog signal digitally using only one bit. Draw neat labelled diagram for it and explain the process in brief.	**05**
ii)	What should be the value of sampling frequency for reproducing analog signal back without any distortion? What will happen if this value not followed? Explain the cases with neat diagram.	**05**
iii)	Identify the errors, if any in the following figure and redraw the figure with the appropriate labelling at point A, B and C. Mention one application of this method.  	**05**
Q.2	Attempt any **one**	**10**
i)	Draw the labelled neat block diagram of generating frequency modulated signal from phase modulated signal. Explain the working in detail. Mention any one advantage of this method.	**10**
ii)	**A.** Explain the working of JFET based reactance FM modulator with neat diagram. **B.** Explain the process of demodulating FM signal using Foster-seeley discriminator with neat phasor diagrams.	**05** **05**
Q.3	Attempt any **one**	**10**
i)	Draw basic block diagram to explain principle of SSB modulation. Why such modulation is advantageous? For SSB transmitter using filter method if, LF carrier frequency f_{LF} =190 KHz, MF carrier frequency f_{MF}=5 MHz, HF carrier f_{HF}=35 MHz, an audio input spectrum f_m =0 KHz to 3KHz. Sketch the frequency spectrum for the following points: (a) BPF1 out, BPF2 out, BPF3 out. (b) Balanced Modulator 1 out, Balanced Modulator 2 out, Balanced Modulator 3 out.	

ii) With reference to Amplitude modulation answer the following:

- Mathematically prove that AM signal has carrier signal, upper and lower sideband signal.
- Mention the conditions of amplitudes of the modulating signal and carrier signal for successful information transmission and information loss.
- How power is distributed in conventional AM signal? Mention formulas
- What is difference between high level and low level modulation?

Q.4 Attempt the following — **10**

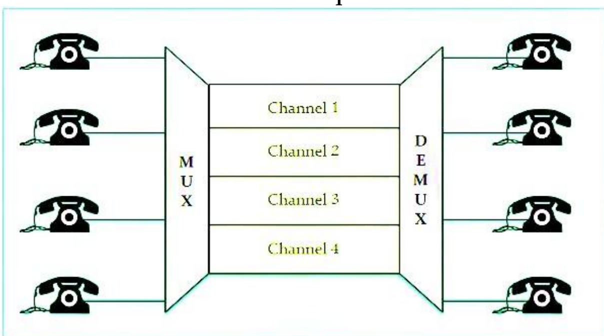
A. Define noise figure and noise factor with its formula. Why these parameters are important for communication system?

B. Justify the name correlated and uncorrelated noise. Give one example of each.

C. “Jack” has certain analog information to be sent to “Jill” at destination. Draw the system block diagram for transmitting this information.

D. Explain concept of modulation with neat diagram.

E. In the figure below if each channel occupies bandwidth of 105 KHz and guard band of 9 KHz is required to prevent interference between them, what is minimum bandwidth required for the link?



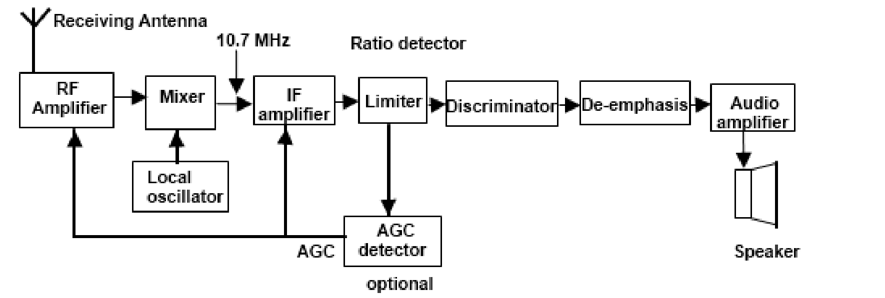
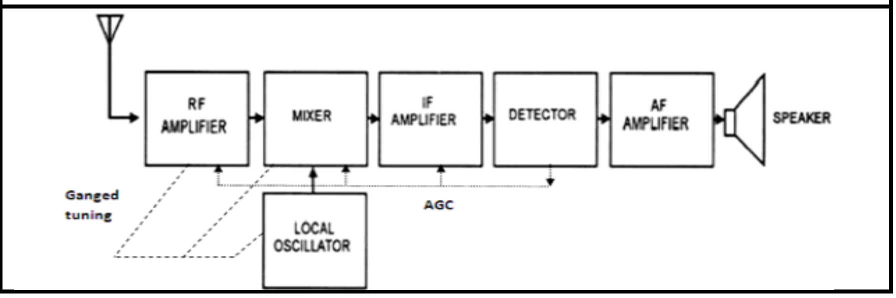
Q.5 Attempt the following — **10**

A. For an AM broadcast band super heterodyne radio receiver with IF, RF and LO frequencies of 455 KHz, 600 KHz and 1055 KHz respective, determine (i) Image frequency (ii) IFRR for pre-selector of Q=100.

B. Define the term sensitivity and selectivity for radio receiver.

C. In receiver ________ AGC waits for the signal to reach certain signal strength before applying gain adjustments, while ________AGC adjusts gain immediately based on the current signal strength.

D. Name the following figure A and B. What is major difference between them?

Fig. A	
Fig. B	

E. An FM wave is given by, $s(t) = 20\cos(8\pi \times 10^6 t + 9\sin(2\pi \times 10^3 t))$. Calculate the frequency deviation, highest and lowest frequencies attained by modulated signal.